

# Ship Auto Way Maps Points Tracking

Dokumentasi sistem **Ship Auto Way Maps Points Tracking** â€” kontrol kapal model ESP32-S3 dengan waypoint, telemetry real-time, mini PC rudder, dan dashboard PySide6.

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**Last update:** 2026-08

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## Daftar isi

1. [Ringkasan sistem](#)
2. [Komponen & path proyek](#)
3. [Arsitektur & alur data](#)
4. [Remote-Side-05](#)
5. [User-Side-05](#)
6. [Dashboard beta 1.5](#)
7. [Mini PC â€” Cpp\\_ReadWriteSerial-1.0](#)
8. [Protokol serial \(Dashboard â†” User-Side\)](#)
9. [Protokol ESP-NOW](#)
10. [Telemetry 24 kolom](#)
11. [Algoritma auto track](#)
12. [Build, upload & menjalankan](#)
13. [Prosedur uji lapangan](#)
14. [Troubleshooting](#)

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## 1. Ringkasan sistem

Sistem ini memungkinkan:

- **Monitoring live** posisi, heading, rudder, RPM, baterai, status Mini PC
- **Perencanaan waypoint** di peta (Home + hingga 10 waypoint navigasi)
- **Pengiriman waypoint** ke kapal via ESP-NOW (0xA1)
- **Kontrol auto alg 2** â€” rudder dari mini PC (timestamp, result)
- **Kontrol auto alg 1** â€” waypoint + PD (opsional, compile-time)
- **Shutdown mini PC** dari dashboard (ESP-NOW 0xA2, tanpa Wi-Fi laptopâ†”mini PC)
- **Analisis log CSV** â€” replay data dengan peta dan plot

Alur end-to-end:

```
Dashboard beta 1.5 (PySide6) â†” USB serial 115200 â†” User-Side-05
(ESP-Now_ESP32-S3_User-Side-05) â†” ESP-NOW peer-to-peer â†” Remote-Side-05
(ESP-Now_ESP32-S3_Remote-Side-05) â†” di kapal â†” USB Serial 115200 â†” Mini PC â†”
Cpp_ReadWriteSerial-1.0
```

## 2. Komponen & path proyek

| Komponen    | Path (dari root repo)                                                |
|-------------|----------------------------------------------------------------------|
| Dashboard   | Pythonfile/Way_Points_Tracking/Local Monitor Dashboard-beta1.5.py    |
| User-Side   | PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_User-Side            |
| Remote-Side | PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_Remote-Side          |
| Mini PC     | Cpp_Files/Cpp_ReadWriteSerial-1.0/                                   |
| Dokumen ini | Pythonfile/Way_Points_Tracking/Ship Auto Way Maps Points Tracking.md |

Versi sebelumnya (referensi): User/Remote-04, Dashboard beta 1.4, Cpp\_ReadWriteSerial (tanpa shutdown).

## 3. Arsitektur & alur data

### 3.1 Telemetry (kapal â†’ PC)

1. Remote baca sensor ~10 Hz â†’ struct DataToSend (64 byte, 24 field)
2. ESP-NOW ke User-Side
3. User-Side cetak CSV 24 kolom ke USB
4. Dashboard parse â†’ Live / log / plot

### 3.2 Waypoint (PC â†’ kapal â†’ mini PC)

1. Dashboard: \$WPSET, ...
2. User-Side â†’ ESP-NOW 0xA1 â†’ Remote simpan RAM + cetak [WP] ...
3. User-Side balas \$WACK, OK / \$WACK, ERR, ...
4. Mini PC (--print all|wp) menampilkan [WP]

### 3.3 Rudder mini PC (auto alg 2)

1. Remote (CH6 auto) kirim CSV 8 kolom ke mini PC
2. Mini PC kirim \$HB (~1 Hz) + timestamp, result
3. Remote set mini\_pc\_link dari heartbeat; pakai result sebagai offset rudder

### 3.4 Shutdown mini PC

1. Dashboard tombol **Shutdown** (hanya jika Mini PC CONNECTED)
2. \$SHUTDOWN â†’ User â†’ ESP-NOW 0xA2 â†’ Remote â†’ Serial \$SHUTDOWN
3. Cpp\_ReadWriteSerial-1.0 jalankan shutdown /s /t 5
4. User balas \$SACK, OK (forward ESP-NOW sukses â€” bukan konfirmasi OS mati)

## 4. Remote-Side-05

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**Proyek:** ESP-Now\_ESP32-S3\_Remote-Side-05 (dari Remote-Side-04)

- Sensor, actuator, RC PPM, waypoint RAM, telemetry ESP-NOW 24 kolom
- USB Serial ke mini PC: CSV 8 kolom, [WP], \$SHUTDOWN; terima \$HB + timestamp, result
- Default #define AUTO\_TRACK\_ALG 2 (mini PC)

Detail: PlatformIO/.../Remote-Side-05/src/README.md

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## 5. User-Side-05

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**Proyek:** ESP-Now\_ESP32-S3\_User-Side-05 (dari User-Side-04)

- Gateway USB â€” ESP-NOW
- Forward \$WPSET â†’ 0xA1, \$SHUTDOWN â†’ 0xA2
- CSV 24 kolom ke dashboard

Detail: PlatformIO/.../User-Side-05/src/README.md

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## 6. Dashboard beta 1.5

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**File:** Local Monitor Dashboard-beta1.5.py (dari beta 1.4)

- Live: mode, Mini PC CONNECTED/DISCONNECTED, warning auto tanpa mini PC
  - Tombol **Shutdown** sebelah status Mini PC (enable jika Connect + mini\_pc\_link=1)
  - Map Points: Home + waypoints, **Send Way Points** (\$WPSET / \$WACK)
  - Logging & Analyze CSV 24 kolom
- 

## 7. Mini PC â€” Cpp\_ReadWriteSerial-1.0

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**Path:** Cpp\_Files/Cpp\_ReadWriteSerial-1.0/ (dari Cpp\_ReadWriteSerial)

| Fitur       | Keterangan                                                     |
|-------------|----------------------------------------------------------------|
| \$HB        | Heartbeat ke Remote ~1 Hz                                      |
| CSV 8 kolom | Baca + hitung rudder (--rudder-mode) + tulis timestamp, result |
| [WP]        | Print ke stdout (--print all\wp)                               |
| \$SHUTDOWN  | Matikan OS Windows/Linux                                       |
| --print     | all / csv / wp                                                 |

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## 8. Protokol serial (Dashboard â†” User-Side)

| Arah        | Format                                        |
|-------------|-----------------------------------------------|
| User â†’ PC | CSV 24 kolom telemetry                        |
| PC â†’ User | \$WPSET, <home_lat>, <home_lon>, <count>, ... |
| User â†’ PC | \$WACK, OK / \$WACK, ERR, <reason>            |
| PC â†’ User | \$SHUTDOWN                                    |
| User â†’ PC | \$SACK, OK / \$SACK, ERR, <reason>            |

Baud: **115200**.

## 9. Protokol ESP-NOW

| msg_type    | Payload                     | Arah            | Fungsi                               |
|-------------|-----------------------------|-----------------|--------------------------------------|
| (telemetry) | DatatoSend 64 B             | Remote â†’ User | Telemetry 24 field                   |
| 0xA1        | waypoints_payload<br>~180 B | User â†’ Remote | Waypoint + home                      |
| 0xA2        | pc_command_payload 4<br>B   | User â†’ Remote | Perintah mini PC (cmd=1<br>shutdown) |

**Catatan:** 0xA2 di versi 05 = shutdown mini PC (bukan tuning NVS dokumen lama).

## 10. Telemetry 24 kolom

Urutan sama di Remote DatatoSend, User CSV, dan dashboard:

1 timestamp, 2 lat, 3 lon, 4 speedMps—100, 5â€”6 servo—100, 7 yaw—100,  
8 hdg\_sp—100, 9 hdg\_err—100, 10 rudder\_cmd—100, 11 track\_wp\_index,  
12 distance\_to\_wp—10, 13â€”18 IMU—100, 19â€”20 RPM, 21â€”22 battery—100,  
23 mode\_auto, **24 mini\_pc\_link**

mode\_auto: 0 Manual, 1 Auto PD, 2 Auto Mini PC.

## 11. Algoritma auto track

Dipilih compile-time di Remote (AUTO\_TRACK\_ALG):

| Nilai | Perilaku                       |
|-------|--------------------------------|
| 1     | Waypoint haversine + PD rudder |



[illegible]